

[P1] Fifteen cards: the histogram route

Mohamed A. Osman — ORCID 0009-0004-5912-999X

Licence: CC BY 4.0. Research draft.

Abstract

No pairwise intersecting 6-uniform family of fifteen cards has transversal number six. Degree bounds force every symbol degree into $\{2, 3, 4\}$; an exhaustive finite lemma on edge covers of K_7 supplies the two hardest degree exclusions; a weighted counting inequality reduces the possible degree histograms to thirteen; and the thirteen are eliminated by incidence arguments. Hence $g(6) \geq 16$.

This route is superseded by [P2], which proves a stronger lemma and reaches the same conclusion in two lines. It is kept because the K_7 lemma and the weighted inequality were developed here, and because it is an independent second argument for the same statement.

Computer-assisted; not independently reviewed. No priority claimed. See the note at the end.

1. Setting

Let H be an indexed pairwise intersecting family of fifteen six-element cards with $\tau(H) = 6$. Repeated cards are allowed at the start, so that a smaller counterexample could be padded up to fifteen without changing its transversal number; §2 shows repeats are impossible anyway, which is what makes the padding legitimate.

For a symbol x , $d(x)$ is its degree, the number of cards containing it. Once degrees are confined to $\{2, 3, 4\}$, write n_2, n_3, n_4 for how many symbols have each degree, and set $t = n_4$. A symbol of degree four is called **high**.

For a card A : q_A is the number of high symbols in A , and ε_A the number of degree-two symbols. Let a_j count the cards with $q_A = j$.

A card with $q_A = 2$ is **small**; a card with $q_A = 6$ is **full**.

2. Degree bounds

Degree one is impossible. Delete the symbol from its only card. The other five symbols of that card meet every other card, since every other card meets this one and cannot do so through a symbol it does not contain. That is a five-cover.

Degree at least seven is impossible. Choose the symbol; at most eight cards remain; pair them and take one shared symbol per pair. Total $1 + 4 = 5$.

Degree six is impossible. Choose the symbol x ; nine cards remain. Suppose every symbol has degree at most two among those nine. Each of the $\binom{9}{2} = 36$ pairs needs a shared symbol, a symbol of degree two serves exactly one pair, and the nine cards hold only $9 \times 6 = 54$ symbol slots, so at most $54/2 = 27$ pairs can be served. Since $27 < 36$, some symbol y has degree at least three among the nine. Then x and y cover at least nine cards, at most six remain, and three more symbols finish them in pairs: $1 + 1 + 3 = 5$.

Lemma 1. *Any pairwise intersecting 6-uniform family of ten cards with maximum symbol degree at most three has a four-cover.*

Proof. Take a degree-three symbol and delete its three cards. If the remaining seven contain a symbol of residual degree at least three, that symbol plus two pairwise- intersection symbols finish the ten, for four in total. Otherwise every symbol of those seven has residual degree exactly two and every pair of the seven meets in exactly one symbol: the seven cards are the vertices of a K_7 and their 21 shared symbols are its edges. Each of the three deleted cards, with the chosen symbol removed, induces an edge cover of that K_7 of size at most five. Lemma 2 below gives a four-edge cover meeting each of them, which is then a four-cover of all ten. $\square$

Degree five is impossible. If $d(x) = 5$, the remaining ten cards cannot have a residual degree-four symbol (choose it and pair the remaining six: $1 + 1 + 3 = 5$). So their maximum degree is at most three, Lemma 1 gives a four-cover, and with x that is five.

Hence every degree lies in $\{2, 3, 4\}$.

No card holds two degree-two symbols. Suppose card C holds degree-two symbols x and y , appearing elsewhere in cards D_x and D_y . Any card other than C , D_x , D_y must meet C , and it cannot do so through x or y , so it meets one of the other four symbols of C . Those four therefore cover everything except possibly D_x and D_y . If those two are distinct they share a symbol; if they coincide, take x . Either way one more symbol finishes, for a total of five. So $\varepsilon_A \in \{0, 1\}$ and $0 \leq n_2 \leq 7$.

Local counts. Counting intersections of a card A with the other fourteen,

$$\sum_{B \neq A} |A \cap B| = 12 + q_A - \varepsilon_A \geq 14,$$

so $q_A \geq 2 + \varepsilon_A$, and in particular $q_A \in \{2, 3, 4, 5, 6\}$. Subtracting fourteen,

$$\sum_{B \neq A} (|A \cap B| - 1) = q_A - \varepsilon_A - 2 \leq 4.$$

A repeated card would contribute 5 to that sum on its own, so repeats are impossible and the padding convention is justified.

3. The finite K_7 lemma

Lemma 2 (computer-assisted). *Given any four edge covers of K_7 , each of size at most five, there is a four-edge cover of K_7 meeting each of the four in an edge. Repetitions among the four inputs are allowed.*

It is enough to test inclusion-minimal inputs. A minimal edge cover is a spanning star forest, and on seven vertices with at most five edges the component patterns are exactly $(3, 2, 2)$, $(5, 2)$ and $(4, 3)$: there are 840 labelled minimal covers — 315 of type $P_3 + 2K_2$, 105 of type $K_{1,4} + K_2$, 420 of type $K_{1,3} + K_{1,2}$ — and 315 four-edge covers.

`code/verify_k7.py` fixes one representative of each of the three isomorphism types as the first input and certifies every choice of the remaining three. Each representative accounts for $\binom{839}{3} = 98,079,939$ quadruples, so 294,239,817 in total, with no counterexample. The method is exact: for each fixed

triple it intersects, over all outputs meeting that triple, the bitsets of inputs each output misses; an empty intersection certifies every fourth input at once. The script also compares that batch method against literal enumeration on nine bounded subcases.

This is a finite computer check, not a claimed hand proof.

4. High symbols and overlap excess

Lemma 3. *Every two high symbols occur together in at least one card.*

Proof. Suppose high symbols x, y have disjoint supports; delete their eight cards. If the remaining seven hold a symbol of residual degree at least three, then x, y , that symbol and two pairwise-intersection symbols give a five-cover. Otherwise the seven carry the K_7 structure of Lemma 1. Each of the four cards containing y , with y removed, induces an edge cover of that K_7 of size at most five. Lemma 2 gives a four-edge cover meeting all four; those four symbols cover the seven residual cards (they are an edge cover) and also the four cards of y (each is met in an edge). With x that is a five-cover. $\square$

For high symbols x, y let λ_{xy} be the number of cards holding both. Lemma 3 gives $\lambda_{xy} \geq 1$; and $\lambda_{xy} = 4$ would make the supports identical, allowing one of the two to be deleted from a card and leaving a five-cover, so $1 \leq \lambda_{xy} \leq 3$.

Define the total excess

$$e = \sum_{x < y} (\lambda_{xy} - 1) = \sum_A \binom{q_A}{2} - \binom{t}{2}.$$

Since a high symbol meets every other high symbol,

$$\sum_{A \ni x} (q_A - 1) \geq t - 1. \quad (\text{P})$$

5. The weighted repetition inequality

Inside a card A , build a graph on its q_A high symbols, joining a pair when $\lambda \geq 2$. Let r_A be its number of edges and

$$w_A = \sum_{\{x,y\} \subseteq A} (\lambda_{xy} - 1).$$

That graph has no independent four-set: four high symbols with no repeated meeting cover exactly $1 + 4 \cdot 3 = 13$ cards, and one shared symbol covers the remaining two, which is a five-cover. A graph on q vertices with r edges has an independent set of size at least $q - r$, so

$$r_A \geq \max(q_A - 3, 0), \quad w_A \geq r_A.$$

For a full card every symbol is high, so $w_A = \sum_{B \neq A} \binom{|A \cap B|}{2} \geq \sum_{B \neq A} (|A \cap B| - 1) = 4$, giving $w_A + 2r_A \geq 10$ there. A pair with $\lambda = 2$ contributes 6 to $\sum_A (w_A + 2r_A)$ and a pair with $\lambda = 3$ contributes 12, so

$$6e = \sum_A (w_A + 2r_A) \geq 3(a_4 + 2a_5 + 3a_6) + a_6. \quad (\text{W})$$

Equivalently, in the a_j alone,

$$6a_2 + 18a_3 + 33a_4 + 54a_5 + 80a_6 \geq 6 \binom{t}{2}. \quad (\text{S})$$

Equality structure, used in §7. If (W) is tight then every full card has $r_A = 3$ and $w_A = 4$; its three repeated pairs form a perfect matching, since two edges sharing an endpoint would leave a vertex cover of size two and hence an independent four-set; and exactly one of those three pairs has $\lambda = 3$. Every $q_A = 5$ card then has $r_A = w_A = 2$, and cards with $q_A \in \{2, 3\}$ have $r_A = w_A = 0$. So in the equality cases a $\lambda = 3$ pair occurs only inside full cards.

6. The thirteen histograms

The necessary numerical conditions are

- $a_2 + a_3 + a_4 + a_5 + a_6 = 15$;
- $\sum_j j a_j = 4t$;
- $2n_2 + 3n_3 + 4t = 90$ with $0 \leq n_2 \leq 7$ and $n_3 \geq 0$;
- $a_3 + a_4 + a_5 \geq 2n_2$, because each of the $2n_2$ cards holding a degree-two symbol has $q_A \in \{3, 4, 5\}$;
- inequality (S).

`code/verify_histograms.py` enumerates these exactly, with no assumed bound on t and no assumed bound on a_2 , and leaves thirteen rows.

(n_2, n_3, n_4)	$(a_2, a_3, a_4, a_5, a_6)$
(0, 10, 15)	(6, 2, 0, 0, 7)
(0, 10, 15)	(7, 0, 0, 2, 6)
(0, 10, 15)	(7, 0, 1, 0, 7)
(1, 12, 13)	(7, 3, 0, 1, 4)
(1, 12, 13)	(8, 1, 0, 3, 3)
(1, 12, 13)	(8, 1, 1, 1, 4)
(1, 12, 13)	(8, 2, 0, 0, 5)
(1, 12, 13)	(9, 0, 0, 2, 4)
(0, 14, 12)	(8, 3, 0, 1, 3)
(0, 14, 12)	(9, 1, 1, 1, 3)
(0, 14, 12)	(9, 2, 0, 0, 4)
(0, 14, 12)	(10, 0, 0, 2, 3)
(0, 14, 12)	(10, 0, 1, 0, 4)

Passing these conditions is necessary, not sufficient: none of the thirteen is claimed to be realizable.

7. Eliminating the thirteen

Throughout, (P) is the tool: a high symbol in two small cards has at most $1 + 1 + 5 + 5 = 12$ high meetings, so at $t = 15$ (which needs 14) it cannot occur in two small cards at all.

$t = 15$

(6, 2, 0, 0, 7). Twelve high symbols lie in small cards and three do not. A high symbol in a $q = 3$ card cannot also lie in a small card, since $1 + 2 + 5 + 5 = 13 < 14$. So both $q = 3$ cards contain exactly the same three absent high symbols. Their intersection then has size at least three, contributing at least two to the local excess, while a $q = 3$ card has excess exactly one.

(7, 0, 1, 0, 7). Fourteen high symbols lie in small cards and one does not. A high symbol lying in both the $q = 4$ card and a small card has at most $1 + 3 + 5 + 5 = 14$ high meetings, so it has no repeated partner. At least three of the four high symbols of the $q = 4$ card are of that kind, so its repeated-pair graph has no edge, contradicting $r_A \geq 1$.

(7, 0, 0, 2, 6). Here $e = 12$ and (W) is tight. Each of the six full cards carries exactly one $\lambda = 3$ pair and such pairs occur only in full cards, so there are exactly two of them, each on three full cards. The three full cards on a fixed $\lambda = 3$ pair $\{x, y\}$ meet pairwise exactly in that pair — a third common symbol would make a triangle in a graph that must be a matching — so their union holds 14 high symbols. If x also lies in a small card $\{x, z, \dots\}$ then z lies outside that union, and its high-meeting total $5 + 5 + 5 + 1 = 16$ has exactly two surplus meetings, both already used by y . If y also lies in a small card, its partner there must be that same unique outside symbol, which would put that symbol in two small cards. So every $\lambda = 3$ pair contains the unique high symbol absent from all seven small cards. Both pairs would then share that symbol; their triples of full cards are disjoint, because a full card's repeated pairs form a matching; and the shared symbol would lie in six cards, contradicting degree four.

$t = 13$

First, $a_2 \leq 7$. By (P) a high symbol in two small cards has its other two appearances in full cards, since $1 + 1 + 4 + 5 = 11 < 12$; hence a high symbol in a card that is neither full nor small lies in at most one small card. The unique degree-two symbol lies in two cards with $q = k \in \{3, 4, 5\}$ and in no small card. Such a card meets at most $k + 2(5 - k) = 10 - k \leq 7$ small cards. So $a_2 \leq 7$, which removes the four rows with $a_2 \in \{8, 9\}$.

(7, 3, 0, 1, 4). Let h count high symbols lying in two small cards. The numbers lying in two, one and zero small cards are $h, 14 - 2h, h - 1$. A double-small high symbol has exactly $1 + 1 + 5 + 5 = 12$ high meetings, so no repeated partner, and its two full cards meet only in it.

For a full card write b_i for the number of its high symbols lying in i small cards. It meets each of the seven small cards exactly once, so $2b_2 + b_1 = 7$, and $b_2 + b_1 + b_0 = 6$, hence $b_2 = b_0 + 1$. Summing over the four full cards, the zero-small high symbols have $2h - 4$ appearances in full cards. Each zero-small high symbol must appear in some full card, since without one its meeting total is at most $2 + 2 + 2 + 4 = 10 < 12$. Hence $2h - 4 \geq h - 1$, i.e. $h \geq 3$.

The h double-small symbols mark distinct pairs of full cards, each marked intersection having size one. If $h \geq 5$ then five of the six pairs have intersection one and the last at most five, so the pairwise intersections sum to at most 10; but four sets of size six inside thirteen symbols force that sum to be at least $24 - 13 = 11$. So $h \in \{3, 4\}$.

A zero-small high symbol lying in one full card must also lie in the $q = 5$ card F and in two of the three $q = 3$ cards; its meeting total is $5 + 4 + 2 + 2 = 13$, allowing exactly one repeat.

Case $h = 3$. Two zero-small symbols u, v , each in exactly one full card, both in F and each in two of the three $q = 3$ cards, so they share F and at least one $q = 3$ card: that uses the single repeat each is allowed. So they lie in different full cards and repeat nothing else. The three marked pairs give the four full cards degrees 1, 2, 2, 1, a path $A - B - C - D$, with $u \in B$ and $v \in C$. The eight single-small high symbols have 16 appearances in full cards, and none can lie in three full cards, since every triple of $\{A, B, C, D\}$ contains a marked pair; so each lies in exactly two. The allowed pairs AC, AD, BD then carry 3, 2, 3 of them. But F holds u, v and three further high symbols, which cannot be double-small and cannot lie in B or C without repeating a meeting with u or v ; they would all have to come from the AD group, which has two members.

Case $h = 4$. Two pairs of full cards are unmarked. Their excess over one must total at least five, since the pairwise intersections sum to at least 11; and they cannot share a full card, whose total intersection excess with all others is only four. So they are disjoint, say AB and CD , and the marked pairs form a four-cycle. Each full card then holds two double-small symbols, one zero-small symbol and three single-small symbols. There are three zero-small symbols with four full-card appearances, so one, w , lies in A and B , and the others u, v lie in C and D respectively. The six single-small symbols cannot lie in three full cards and have twelve appearances, so each lies in two: three shared by A, B and three by C, D . Thus w already repeats meetings with three symbols. If its other two cards were both $q = 3$, its meeting total would be $5 + 5 + 2 + 2 = 14$, allowing only two surplus meetings. So w lies in F and one $q = 3$ card. But u and v each lie in F and in two of the three $q = 3$ cards, and each has used its single allowed repeat on the other; their $q = 3$ supports overlap in exactly one card and together cover all three. So whichever $q = 3$ card holds w also holds u or v , repeating that meeting on top of the u, v repeat.

$t = 12$

A high symbol in a $q = 3$ or $q = 4$ card cannot lie in two small cards, since $1 + 1 + 3 + 5 = 10 < 11$.

(9, 1, 1, 1, 3) and (10, 0, 1, 0, 4). The $q = 4$ card holds four high symbols, each in at most one small card, and two degree-three symbols, each in at most two; so it meets at most $4 + 2 \cdot 2 = 8$ small cards, fewer than the nine or ten required.

(9, 2, 0, 0, 4). A $q = 3$ card meets at most $3 + 3 \cdot 2 = 9$ small cards, so equality is forced everywhere. The two $q = 3$ cards cannot share a degree-three symbol (not enough remaining appearances) nor a high symbol (its meeting total would be at most $1 + 2 + 2 + 5 = 10 < 11$), so they would be disjoint.

(10, 0, 0, 2, 3). Each $q = 5$ card must hold at least three double-small high symbols to meet all ten small cards; otherwise its five high symbols and one degree-three symbol reach at most $5 + 2 + 2 = 9$. Each double-small high symbol in it has its fourth appearance in a full card and meeting total $1 + 1 + 4 + 5 = 11$, so that full card meets the $q = 5$ card only in it. With three full cards there are exactly three such symbols, one per full card; the other two high symbols lie in one small card each and the degree-three symbol in two. The two $q = 5$ cards cannot share their degree-three symbol or a double-small symbol, and a shared single-small high symbol would add a second symbol to an intersection already forced to be a singleton. So they would be disjoint.

(8, 3, 0, 1, 3). Here $e = 6$ and (W) is tight, so each full card has a repeated-pair matching of size three with exactly one $\lambda = 3$ pair, and non-full cards hold no $\lambda = 3$ pair. So there is exactly one

such pair, shared by all three full cards. As in $t = 15$, two full cards cannot share a third symbol, so their union holds $2 + 3 \cdot 4 = 14$ high symbols, while only twelve exist.

8. Conclusion

Theorem 1. *No pairwise intersecting 6-uniform family of fifteen cards has transversal number six. Since repeats were shown impossible after padding was allowed, the same holds for every family of at most fifteen cards. Hence $g(6) \geq 16$.*

9. Verification

claim	script
Lemma 2, all 294,239,817 quadruples	<code>code/verify_k7.py</code>
the thirteen histograms	<code>code/verify_histograms.py</code>

The eliminations of §7 are hand arguments. They are not machine-checked.

This work was prepared with AI assistance (ChatGPT, OpenAI; Claude, Anthropic), used for algebraic derivation, for drafting and rewriting code and text, for running and re-running the computations, and for auditing the papers against their own scripts. The same tools were used in the review, so the review carries the same caveat. All statements were checked by the author, who is responsible for them. No priority is claimed and the result has not been independently reviewed by a human or a proof assistant; the repository README sets out the division of labour and the open obligations in full.

References

1. J. Barát, *Intersecting and 2-intersecting hypergraphs with maximal covering number: the Erdős–Lovász theme revisited*, arXiv:2011.04444.
2. V. Sivashankar, *An Improved Lower Bound for the Erdős–Lovász Cover Number Problem*, arXiv:2606.24878v2.